EV MARKET ANALYSIS DASHBOARD

Project Documentation & Presentation Companion

An Exploratory Data Visualization Project Using Tableau Public 2026.3

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Source file: export.csv
Final workbook format: Tableau packaged workbook (.twbx)

Based on the existing project report and dashboard validation notes. Screenshot-observed values are marked for verification and should not be treated as final statistics until checked in Tableau.

Executive Summary

This project uses Tableau Public 2026.3 to turn an electric-vehicle registration CSV into a dashboard for exploratory analysis. The core dashboard combines four views: top manufacturers, vehicle-type distribution, registrations by model year, and geographic distribution. The main measure is a distinct count of **DOL Vehicle ID**, rather than a sum of identifier values.

The reviewed screenshots suggest Tesla is the largest manufacturer category, BEV records outnumber PHEV records in the displayed data, and recent model years rise to a visible peak around 2023. These are observations, not verified final statistics. Counts, dataset coverage, and map details should be checked in Tableau before formal submission.

1. Introduction

EV registration data can help users explore manufacturer distribution, vehicle-type composition, model-year patterns, and geographic concentration. A raw table is difficult to scan, so this project summarizes selected fields in visual form. The supplied **export.csv** dataset was used in Tableau Public 2026.3; the final Tableau deliverable is a packaged workbook (.twbx).

2. Problem Statement

The raw dataset contains multiple dimensions and many records. Row-by-row inspection does not immediately reveal manufacturer rankings, vehicle-type distribution, model-year patterns, or geographic concentration. The project addresses this by transforming selected fields into visualizations that can be interpreted together.

3. Objectives

- Compare manufacturers using distinct registered vehicle IDs.
- Compare registrations by EV type.
- Visualize distinct vehicle counts by model year.
- Display geographic patterns using county/state fields.
- Combine views into one dashboard and validate the packaged workbook.

4. Scope and Boundaries

This is an exploratory visualization of the supplied records. It does not establish causal relationships or forecast future adoption. It should not be presented as a complete view of the entire EV market unless the source dataset's temporal and geographic coverage is verified. The reviewed map appears to focus on Washington State counties.

5. Dataset Description

Field	Purpose
DOL Vehicle ID	Vehicle identifier; distinct count used in charts.
Make	Manufacturer category.
Model Year	Year dimension for the trend.
Electric Vehicle Type	Vehicle type, including BEV/PHEV categories.
County / City / State	Geographic dimensions for the map.
Postal Code	Additional location field.
Model	Vehicle model information for possible detailed analysis.

Not confirmed in reviewed screenshots: dataset source URL, extraction date, exact row count, missing-value count, and duplicate-record count. Check the original dataset/documentation rather than estimating these values.

6. Tools and Technologies

- Tableau Public 2026.3 — data connection, worksheets, visualization, dashboard assembly.
- CSV — source data format.
- Tableau packaged workbook (.twbx) — final Tableau deliverable.
- Screen capture utility — optional screenshots for review.

7. Methodology

1. Connect export.csv in Tableau Public and inspect the fields.
2. Check field types and choose dimensions and measures.
3. Build manufacturer and vehicle-type charts using COUNTD(DOL Vehicle ID).
4. Build a model-year line chart and a geographic map using appropriate geographic roles.
5. Assemble worksheets on one dashboard and refine titles/layout.
6. Review aggregations and filters; save as .twbx, close, reopen, and check connectivity.

Core calculation: COUNTD([DOL Vehicle ID]) — distinct count of vehicle identifiers.

8. Dashboard Visualizations

8.1 Top 10 EV Manufacturers by Registered Vehicles

A horizontal bar chart ranks manufacturers by distinct count of DOL Vehicle ID. The Make field provides the category and the distinct-count measure provides the bar length. Tesla appears as the longest bar in the reviewed screenshot. Approximate screenshot-observed labels include Tesla 122,291, Ford 15,997, Chevrolet 20,218, Audi 6,053, Hyundai 12,581, and Porsche 2,170. Verify the exact values and ordering in Tableau before quoting them.

8.2 Vehicle Distribution by Type

This horizontal bar chart compares categories in Electric Vehicle Type. Screenshot-observed approximate values were BEV 241,840 and PHEV 58,302. The view suggests more BEV than PHEV records in the displayed data. Confirm values in the worksheet or tooltips.

8.3 EV Registrations by Model Year

The line chart uses Model Year on the horizontal axis and distinct vehicle-ID count on the vertical axis. The reviewed view rises substantially in recent model years, with a visible peak around 2023 and lower values afterward in the unfiltered view. A 1999–2025 filter was tested while investigating the later-year tail. Retain it only if supported by source-data completeness or project scope—not merely to make the line smoother.

8.4 Geographic Distribution

The map uses available geographic fields and appears to display counties in Washington State, shaded by the selected registration measure. Darker areas represent comparatively higher values. County names, counts, and exact coverage were not confirmed in screenshots; check map tooltips before naming a leading county.

9. Problems Encountered and Solutions

Incorrect aggregation: summing an identifier is not a meaningful count. **Solution:** use COUNTD(DOL Vehicle ID).

Misleading trend labels: the model-year view initially showed SUM(DOL Vehicle ID). **Solution:** remove incorrect sum labels and retain distinct counts.

Later-year records: the trend extends into years whose completeness needs investigation. **Solution:** verify the source data and justify any year filter.

Unverified chart values: screenshot labels may not reflect the final worksheet state. **Solution:** verify tooltip/worksheet values before final reporting.

Workbook portability: local data paths may fail on another system. **Solution:** save as .twbx, close, reopen, and test data connectivity.

10. Results and Observations

- Tesla is visually the largest manufacturer category in the reviewed screenshot; exact counts require verification.
- BEV appears larger than PHEV in the displayed dataset; exact counts require verification.
- The model-year view shows a recent rise and peak around 2023; later-year completeness needs checking.
- The map supports geographic comparison, but coverage appears Washington-focused.

11. Limitations

- Source URL and extraction date were not confirmed.
- Exact row count and missing/duplicate-record statistics were not confirmed.
- Some numeric values were read from screenshots and require tooltip verification.
- Completeness of the latest model years needs investigation.
- Geographic coverage should not be generalized beyond what the data confirms.
- Results describe the supplied records; they are not causal or predictive.

12. Future Enhancements

1. Add a clear source citation, extraction date, and coverage statement.
2. Verify all counts and document data-cleaning decisions.
3. Add useful filters for year, vehicle type, manufacturer, and geography.
4. Improve tooltips with plain-language definitions and verified values.
5. Annotate notable peaks without implying causation.
6. Consider per-capita/normalized comparisons only if suitable denominator data is available.
7. Add final dashboard screenshots and project documentation to the repository.

13. Conclusion

The EV Market Analysis Dashboard demonstrates how Tableau can convert a flat CSV into coordinated views that make manufacturer comparisons, vehicle-type distribution, model-year patterns, and geographic differences easier to inspect. Using a distinct count of vehicle IDs improves consistency. The most important remaining checks are to verify screenshot-observed counts, confirm dataset provenance and geographic coverage, and justify the model-year filter based on data completeness.

14. Project Links and References

GitHub repository: https://github.com/shivamk26/EV-Market-Analysis-Dashboard
Software: Tableau Public 2026.3
Data file: export.csv
Primary artifact: Tableau packaged workbook (.twbx)

Important note: this document is based on the existing project report and validation notes. It does not independently verify the final workbook's live values. Verify numerical results in Tableau before treating them as final statistics.